

Adrian Morton-Torricella

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EDUCATION

Florida International University

Aug 2023 - May 2027

Bachelor of Science, Computer Science | Minor in Mathematical Sciences

- **Achievements:** Dean's List (Spring 2024, Summer 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- **Coursework:** DS&A, Operating Systems, Net-Centric Computing, Database Management, Intro to Machine Learning, Intro to Deep Learning, Fund. of Quantum Computing, Algorithm Techniques, Software Engineering I/II, Intro to Data Mining

WORK EXPERIENCE

Stemtree of Coral Gables | *Programming Instructor*

Jun 2025 - Present

- Designed and delivered hands-on programming curriculum for students across all skill levels and various programming languages (**Python**, **Java**, **C/C++**), focusing on high-school level enrichment.
- Mentored students through project-based activities that developed critical thinking, problem-solving, and technical proficiency, leading to improved student performance and deepening my own proficiency in many programming languages.

Portable Diagnostic Systems | *Software Engineering Intern*

May 2026 - Present

- Designing and building an **OTA software update deployment system** for the Integrity-1 Analysis System, enabling remote firmware/software rollout across a fleet of units without physical access, eliminating manual reflashing and on-site servicing.

Genuine Labs | *Software Engineering Intern*

Jun 2026 - Present

- Architecting and building an **LLM evaluation framework** from the ground up to support rigorous testing of production AI systems.
- Designed a JSON-driven eval runner capable of executing structured test cases against LLM endpoints, with support for both reference-based grading and **LLM-as-judge** scoring methodologies.
- Engineered a dual-store memory architecture using **mem0** and **Firestore** with test isolation via purge steps, enabling reproducible eval runs in CI environments.
- Integrated **Firebase Admin SDK** for service-account-based authentication, resolving cross-environment auth challenges.
- Collaborated cross-functionally on memory selection logic, model provider migration, and dummy Firebase environment setup for safe dev/test iteration.
- Built logging and failure reporting pipelines to surface actionable signals from eval runs.

PROJECTS

Repro | *1st Place, Microsoft Sponsor Challenge* | *ShellHacks X*

- Won **1st place in the Microsoft sponsor challenge** at ShellHacks X with Repro, a **36-hour**, 4-person build that points **AI agents** at a codebase, reproduces every finding in a sandbox before reasoning about it, and repairs code with proof the fix holds.
- Built the **agentic diagnosis and autonomous repair** lane in a **TypeScript** monorepo, where **Gemini** groups reproduced findings by root cause and edits code only through sandboxed tools, and an adversarial Challenger model writes counter-tests to break each patch.
- Kept detection deterministic with **Semgrep**, **gitleaks**, osv-scanner, and Ruff in network-isolated **Docker** containers, driven by a **Fastify/tRPC** control plane with **MongoDB Atlas** run storage that opens verified fixes as GitHub pull requests.

NoteBud | *INIT Build Spring 2026*

- **Leading** the development of NoteBud, an AI-powered class notebook inspired by **Google NotebookLM** and **Obsidian** built with **Python** and **TypeScript** that will use **retrieval-augmented generation** to answer student questions with citations from their own course materials and notes.
- Developing a full-stack application with a **Next.js** frontend, **FastAPI** backend, **Neo4j** graph database, **Docker** containerization, and **Google Cloud Storage** to manage users, notebooks, notes, and course materials.
- Designing RAG pipeline using **LlamaIndex/LangChain** and **Gemini** to ingest PDFs/slides, chunk and embed content, process through retrieval ranking, groundedness scoring, and then return informed responses with highlighted evidence.

Prenergyze | *INIT Build Fall 2025*

- **Lead** the development of predictive systems built off of machine learning models in **Python** to anticipate electric utility grid load peaks, supplying the smart grid with the means to enable dynamic pricing and efficient resource allocation.
- Developed, trained, and evaluated models built using **PyTorch** including **Catboost (R2 = 0.9300)**, **LightGBM (R2 = 0.9318)**, and **Temporal Fusion Transformer (R2 = 0.9434)** to balance accuracy and interpretability.

Specgate | *Personal*

- Architected a **TypeScript** monorepo (engine, CLI, GitHub Action, shared contracts, control plane) that verifies AI-generated applications against machine-readable specs, emitting per-requirement verdicts with attached request and response evidence.
- Enforced a zero-LLM verdict path so every verdict is reproducible, using **ESLint** import boundaries asserted by tests that read the resolved config rather than the config file, so a reordering that silently drops the rule fails CI.
- Developed a source-first probe with adapters for **Next.js**, **Express**, and **Prisma**, backed by a deterministic HTTP runner with injected clock and identifier sources for cross-request and persisted-state assertions.
- Engineered a three-valued check engine (pass/fail/inconclusive) with **9** distinct unverified reasons to prevent false passes, validated by **~1,890 tests** and a **24-application** corpus of seeded defects with every finding hand-classified to measure the false-positive rate.

SKILLS

- **Languages and Tools:** Python, C/C++, Java, SQL, Arduino, C#, Typescript, Javascript, HTML/CSS, Rust, Linux, Git, Wireshark, Oracle Virtualbox, Fusion360, Onshape, Shapr3D, Postman
- **Libraries and Frameworks:** PyTorch, Docker, pandas, TensorFlow, NumPy, Qiskit, Matplotlib, Seaborn, React, Node.js, Flask, Restful API, LangChain, LlamaIndex, Google ADK, PostgreSQL, FastAPI, Google Cloud, Vector Databases, Apache Parquet, Neo4j